

An elastic network model with $k_{ij} = k(l_{ij})$

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2026-08-26

Abstract

This note sets up an elastic network model of proteins whose goal is to predict how mutations change structure, motions, and stability. In standard ENMs the parameters are fixed by a known structure: each spring's rest length is the distance between its residues in that structure, so modeling a mutant requires supplying the mutant structure, which is itself one of the things to be predicted. Here, instead, we tie spring stiffnesses to rest lengths, $k(l_{ij})$, so the rest lengths alone define the model and the structure follows by minimization. A mutation is then just a perturbation of the rest lengths at a site, and the mutant's structure, motions, and thermodynamic changes (energy, entropy, and free energy) are derived, not assumed. A side effect is that networks can now carry strain and frustration, which real proteins are known to display but standard ENMs exclude by construction. We work out the full formalism and end with a list of open questions.

1 Introduction

A protein can be modelled as an elastic network of nodes, representing amino acids, connected by springs, representing interactions. Such an elastic network model (ENM) is completely specified by the spring force constants k_{ij} and the spring equilibrium lengths l_{ij} . Typically, given a native conformation $\mathbf{r}^e$, the ENM is set up so that $l_{ij} = d_{ij}(\mathbf{r}^e)$ and $k_{ij} = k(d_{ij}(\mathbf{r}^e))$. The structure thus enters twice: it fixes the parameters, and it is the conformation at which the model is relaxed.

We would like to use such a model to predict how a mutation changes a protein: its structure, its motions, its stability. But a mutation changes the parameters, and here the parameters are the structure. To mutate the model one must hand it the mutant structure, which is what we wanted it to predict. The same identification has a second consequence: every spring is relaxed at $\mathbf{r}^e$, so the network carries no strain, and frustration, which proteins are known to display, cannot be represented at all.

Both problems come from tying the parameters to a structure, so we untie them. We take the force constants to be a function of the equilibrium lengths of the springs, $k_{ij} = k(l_{ij})$, rather than of the distances between nodes in a given structure. The model is then determined by the l_{ij} alone. The springs need not be relaxed at the equilibrium conformation, but may be strained, so the network may be frustrated; and a mutation of a site is modelled by perturbing the equilibrium lengths of the springs connected to it, with the mutant structure following from the model rather than being supplied to it.

This note derives the potential energy function, its minimum, the conformational distribution, and thermodynamic properties, first for a general network and then for a reference and its mutant, ending with the mutational changes.

2 The model

2.1 Nodes

A protein is represented by N nodes, one per residue, joined in pairs by springs. A conformation is the set of the node positions,

$$\mathbf{r} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \in \mathbb{R}^{3N}. \quad (1)$$

We write

$$\mathbf{r}_{ij} \equiv \mathbf{r}_j - \mathbf{r}_i \quad (2)$$

for the vector that goes from node i to node j . The distance between the two nodes in conformation $\mathbf{r}$ is the norm of this vector:

$$d_{ij}(\mathbf{r}) = \|\mathbf{r}_{ij}\|. \quad (3)$$

2.2 Springs

Every pair of nodes is joined by a spring. A spring properties are its force constant k_{ij} and its equilibrium length l_{ij} . We assume the force constant of a spring to be determined by its own equilibrium length through a function k ,

$$k_{ij} = k(l_{ij}), \quad k : [0, \infty) \rightarrow [0, \infty). \quad (4)$$

We assume of k only that it is non-increasing, so that long springs are weakly coupled or uncoupled, and that $k(l) = 0$ for l beyond some range, so that springs long enough to represent no contact do not interact.

Since the k_{ij} are not independent parameters, the model is determined by the set of length parameters:

$$\Lambda = \{l_{ij} : 1 \leq i < j \leq N\}, \quad |\Lambda| = \frac{1}{2}N(N-1). \quad (5)$$

This is the one departure from standard practice, and it is worth stating carefully, because in the standard case the departure is invisible. A standard elastic network model is built on a given structure, which it takes as the equilibrium conformation $\mathbf{r}^e$, and assigns each spring the equilibrium length $l_{ij} = d_{ij}(\mathbf{r}^e)$ it already has in that structure. Its force constants are then written as a function of $d_{ij}(\mathbf{r}^e)$ — but since l_{ij} and $d_{ij}(\mathbf{r}^e)$ are equal there, a function of one is a function of the other, and nothing distinguishes $k(l_{ij})$ from $k(d_{ij}(\mathbf{r}^e))$. Here the l_{ij} are left free rather than read off a structure, the two cease to coincide, and (4) states which of them k is a function of: the parameter l_{ij} .

2.3 The potential

The potential energy of a conformation $\mathbf{r}$ for a network with parameters Λ is a sum of harmonic terms, one per spring,

$$V(\mathbf{r}; \Lambda) = \frac{1}{2} \sum_{i < j} k(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij})^2. \quad (6)$$

The sum runs over all $\frac{1}{2}N(N-1)$ springs, and V evaluated at any conformation is the elastic energy stored in them. Every term is non-negative, so $V \geq 0$, and $V(\mathbf{r}; \Lambda) = 0$ requires every spring to sit at its own equilibrium length, $d_{ij}(\mathbf{r}) = l_{ij}$ for every active spring at once. Whether any conformation achieves this is a question about Λ , not a property of the model: for some parameter sets no conformation does, and then $V > 0$ everywhere. Section 3.3 takes this up.

Do we really
sume this?

A spring with $k(l_{ij}) > 0$ contributes to V and is said to be *active*; a spring with $k(l_{ij}) = 0$ contributes nothing. The set of active springs is the network's contact map. Since k is a function of l_{ij} , which springs are active is fixed by Λ alone: the contact map is a property of the parameters, and does not change as the network moves. In a standard model, where $l_{ij} = d_{ij}(\mathbf{r}^e)$, this same set would be read off the structure instead.

3 The minimum of the potential

3.1 The equilibrium condition

To find the conformations that minimise (6) we need the gradient of V , and hence the gradient of the distances. Let

$$\hat{\mathbf{e}}_{ij} \equiv \frac{\mathbf{r}_{ij}}{d_{ij}} \quad (7)$$

be the unit vector along the spring, pointing from i to j ; it satisfies $\hat{\mathbf{e}}_{ij} = -\hat{\mathbf{e}}_{ji}$. Differentiating $d_{ij}^2 = \mathbf{r}_{ij} \cdot \mathbf{r}_{ij}$ with respect to $\mathbf{r}_i$, and using $\partial \mathbf{r}_{ij} / \partial \mathbf{r}_i = -\mathbf{I}$,

$$2 d_{ij} \frac{\partial d_{ij}}{\partial \mathbf{r}_i} = -2 \mathbf{r}_{ij}, \quad \text{so} \quad \frac{\partial d_{ij}}{\partial \mathbf{r}_i} = -\frac{\mathbf{r}_{ij}}{d_{ij}} = -\hat{\mathbf{e}}_{ij}. \quad (8)$$

The derivative with respect to $\mathbf{r}_j$ follows the same way, now with $\partial \mathbf{r}_{ij} / \partial \mathbf{r}_j = +\mathbf{I}$, giving $\partial d_{ij} / \partial \mathbf{r}_j = +\hat{\mathbf{e}}_{ij}$. Moving node i away from node j lengthens the spring, and (8) records that: the distance decreases fastest along $-\hat{\mathbf{e}}_{ij}$, which points from j back towards i .

Differentiating (6) by the chain rule, and noting that $k(l_{ij})$ is a constant with respect to $\mathbf{r}$,

$$\frac{\partial V}{\partial \mathbf{r}_i} = - \sum_{j \neq i} k(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij}) \hat{\mathbf{e}}_{ij}. \quad (9)$$

Each term lies along the line joining the two nodes, so the force it exerts is central: it has no component transverse to the spring. That k carries no derivative is a consequence of (4). Had the force constants been written as a function of the equilibrium distance, $k_{ij} = f(d_{ij}(\mathbf{r}^e))$, they would still be constants with respect to $\mathbf{r}$ and this gradient would be unchanged; what (4) buys is not simplicity here but the freedom, used in Section 6, to vary l_{ij} and have k_{ij} follow.

Given the parameters Λ , an equilibrium conformation $\mathbf{r}^e$ is one at which the gradient vanishes,

$$\boxed{\sum_{j \neq i} k(l_{ij}) (d_{ij}(\mathbf{r}^e) - l_{ij}) \hat{\mathbf{e}}_{ij}^e = \mathbf{0}, \quad i = 1, \dots, N,} \quad (10)$$

where the overall sign of (9) has been dropped, the right-hand side being zero. These are $3N$ scalar equations for the $3N$ unknown coordinates, of which $3N - 6$ are independent: V depends on the conformation only through the distances (3), so it is invariant under the three translations and three rotations, and along those six directions the gradient vanishes identically.

One consequence of that invariance should be recorded now, because it matters later: $\mathbf{r}^e$ is determined only up to a rigid motion.

Separately, and for a different reason, V is also invariant under reflection. This does not follow from invariance under rigid motions — a reflection is not one — but from the same fact as before, that V sees the conformation only through the distances (3), and reflecting a conformation leaves every d_{ij} unchanged. A chiral $\mathbf{r}^e$ therefore has a mirror image of equal energy, and the minimum of V is not unique. Below, $\mathbf{r}^e$ denotes one chosen minimum, and every quantity derived from it is understood relative to that choice.

3.2 Strain

The combination $d_{ij}(\mathbf{r}^e) - l_{ij}$ appearing in (10) is the amount by which a spring departs from its equilibrium length at the equilibrium conformation. It is convenient to name it. Let $\mathbf{r}^e$ be an equilibrium conformation and define, for every spring,

$$s_{ij} \equiv d_{ij}(\mathbf{r}^e) - l_{ij}, \quad (11)$$

its *strain*. A spring with $s_{ij} > 0$ is longer than its equilibrium length and so is stretched, one with $s_{ij} < 0$ is compressed, and one with $s_{ij} = 0$ is relaxed.

In these terms the equilibrium condition (10) reads

$$\sum_{j \neq i} k(l_{ij}) s_{ij} \hat{\mathbf{e}}_{ij}^e = \mathbf{0}, \quad i = 1, \dots, N, \quad (12)$$

so the forces on each node cancel, which does not require the individual s_{ij} to vanish. Substituting (11) into (6) evaluated at $\mathbf{r}^e$ gives the minimum energy

$$V(\mathbf{r}^e; \Lambda) = \frac{1}{2} \sum_{i < j} k(l_{ij}) s_{ij}^2, \quad (13)$$

which depends on the strains only through their squares, and so is unchanged by the sign convention adopted in (11).

3.3 Frustration

We call a network **frustrated** when $V(\mathbf{r}^e; \Lambda) > 0$: no conformation relaxes all of its active springs at once, and the network sits at a compromise in which the forces cancel node by node while individual springs remain strained. By (13) this is the case whenever $s_{ij} \neq 0$ for at least one active spring. A network that is not frustrated has $s_{ij} = 0$ for every active spring, and then $V(\mathbf{r}^e; \Lambda) = 0$.

Whether a given Λ is frustrated is settled by solving (10) for $\mathbf{r}^e$ and evaluating (13). One case can be read off without solving anything. Choose a structure $\mathbf{r}^*$ and set $l_{ij} = d_{ij}(\mathbf{r}^*)$ for every spring. Then every strain vanishes at $\mathbf{r}^*$, so (12) is satisfied term by term, $\mathbf{r}^*$ is an equilibrium conformation, and $V(\mathbf{r}^*; \Lambda) = 0$: the network is unfrustrated by construction. This is what a standard elastic network model does, and it is a statement about how Λ was chosen, not about (10), which takes Λ as given.

Nothing obliges Λ to have been chosen that way, and parameter sets not of this form will in general be frustrated. How the l_{ij} ought to be chosen is a separate question from the one (10) answers, and is not taken up here; Section 9 returns to it. What matters below is that both cases are allowed: nothing in Sections 4 to 7 assumes an unfrustrated network. As will be seen in Section 6, a mutation applied to an unfrustrated network produces a frustrated one.

4 The conformational distribution

4.1 The Boltzmann distribution

At temperature T the conformations are distributed according to

$$\rho(\mathbf{r}) = \frac{e^{-\beta V(\mathbf{r}; \Lambda)}}{Z}, \quad Z = \int e^{-\beta V(\mathbf{r}; \Lambda)} d\mathbf{r}, \quad \beta = \frac{1}{k_B T}, \quad (14)$$

where the integral runs over $\mathbb{R}^{3N}$. This is a configurational integral: the momenta have been left out. They contribute a factor that depends on the masses and on β but not on Λ , so it cancels from every difference between networks computed in Section 7, which is all we use.

4.2 The Hessian

The integral in (14) cannot be evaluated in closed form, because V is not quadratic in $\mathbf{r}$: the distances (3) are nonlinear functions of the coordinates. We therefore expand V about an equilibrium conformation. Writing $\mathbf{u} = \mathbf{r} - \mathbf{r}^e$,

$$V(\mathbf{r}; \Lambda) = V(\mathbf{r}^e; \Lambda) + \left. \frac{\partial V}{\partial \mathbf{r}} \right|_{\mathbf{r}^e} \cdot \mathbf{u} + \frac{1}{2} \mathbf{u}^T \mathbf{K} \mathbf{u} + O(\mathbf{u}^3), \quad \mathbf{K} = \left. \frac{\partial^2 V}{\partial \mathbf{r}^2} \right|_{\mathbf{r}^e}. \quad (15)$$

The linear term vanishes by (10).

To obtain $\mathbf{K}$ we need the second derivatives of the distances. Differentiating $\hat{\mathbf{e}}_{ij} = \mathbf{r}_{ij}/d_{ij}$ with respect to $\mathbf{r}_j$, using the product rule with $\partial \mathbf{r}_{ij}/\partial \mathbf{r}_j = \mathbf{I}$ and $\partial d_{ij}^{-1}/\partial \mathbf{r}_j = -d_{ij}^{-2} \hat{\mathbf{e}}_{ij}^T$,

$$\frac{\partial \hat{\mathbf{e}}_{ij}}{\partial \mathbf{r}_j} = \frac{\mathbf{I}}{d_{ij}} - \frac{\mathbf{r}_{ij} \hat{\mathbf{e}}_{ij}^T}{d_{ij}^2} = \frac{\mathbf{I} - \hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T}{d_{ij}}. \quad (16)$$

Now differentiate the gradient (9) with respect to $\mathbf{r}_j$ for $j \neq i$. Only the single term of the sum with that index depends on $\mathbf{r}_j$, and it is a product of $(d_{ij} - l_{ij})$ and $\hat{\mathbf{e}}_{ij}$, so

$$\mathbf{K}_{ij} = -k(l_{ij}) \left[\frac{\partial (d_{ij} - l_{ij})}{\partial \mathbf{r}_j} \hat{\mathbf{e}}_{ij}^T + (d_{ij} - l_{ij}) \frac{\partial \hat{\mathbf{e}}_{ij}}{\partial \mathbf{r}_j} \right]. \quad (17)$$

Substituting $\partial d_{ij}/\partial \mathbf{r}_j = \hat{\mathbf{e}}_{ij}$ and (16), and then $d_{ij} - l_{ij} = s_{ij}$ from (11),

$$\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T + s_{ij} \frac{\mathbf{I} - \hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T}{d_{ij}} \right], \quad (18)$$

that is, defining $g_{ij} \equiv s_{ij}/d_{ij}(\mathbf{r}^e)$,

$$\boxed{\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T - g_{ij} (\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T - \mathbf{I}) \right], \quad i \neq j.} \quad (19)$$

The diagonal blocks follow from translational invariance. Displacing every node by the same vector $\mathbf{a}$ leaves V unchanged, so $\mathbf{K}$ annihilates that displacement, and summing the blocks in a row gives $\sum_j \mathbf{K}_{ij} \mathbf{a} = \mathbf{0}$ for every $\mathbf{a}$, whence

$$\mathbf{K}_{ii} = -\sum_{j \neq i} \mathbf{K}_{ij}. \quad (20)$$

Inactive springs have $k(l_{ij}) = 0$ and contribute nothing, so $\mathbf{K}$ is $3N \times 3N$ however many springs carry parameters.

4.3 Longitudinal and transverse stiffness

Equation (19) contains two projectors. Writing $\mathbf{P}_\perp = \mathbf{I} - \hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T$ for the projector onto the plane perpendicular to the spring, it may be rearranged as

$$\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T + g_{ij} \mathbf{P}_\perp \right], \quad (21)$$

so a spring carries two stiffnesses rather than one: a longitudinal $k(l_{ij})$ resisting change of the distance, and a transverse $k(l_{ij})g_{ij}$, doubly degenerate, resisting rotation of the spring at fixed distance.

The transverse stiffness is proportional to $g_{ij} = s_{ij}/d_{ij}(\mathbf{r}^e)$, and so vanishes for a relaxed spring and changes sign with the strain. A stretched spring ($s_{ij} > 0$) resists transverse displacement, as a taut string does. A compressed one ($s_{ij} < 0$) has negative transverse stiffness: displacing the two nodes sideways increases their separation, which a compressed spring favours. Note that it is the transverse stiffness that is negative, not $k(l_{ij})$, which is non-negative by (4).

Since $g_{ij} = 0$ for every spring of an unfrustrated network, the transverse term is absent there, and (19) reduces to the rank-one form of a standard elastic network model. Frustration enters the conformational distribution through g_{ij} and through nothing else.

4.4 The Gaussian approximation

Truncating (15) after the quadratic term gives

$$V(\mathbf{r}; \Lambda) \simeq V(\mathbf{r}^e; \Lambda) + \frac{1}{2} \mathbf{u}^\top \mathbf{K} \mathbf{u}, \quad (22)$$

and hence, from (14), a Gaussian distribution of conformations centred on $\mathbf{r}^e$,

$$\rho(\mathbf{r}) \simeq \frac{1}{Z} e^{-\beta V(\mathbf{r}^e; \Lambda)} \exp\left(-\frac{\beta}{2} \mathbf{u}^\top \mathbf{K} \mathbf{u}\right). \quad (23)$$

$\mathbf{K}$ is symmetric, and positive semi-definite because $\mathbf{r}^e$ is a minimum. Let its eigenvalues be λ_n with eigenvectors $\mathbf{q}_n$. At least six eigenvalues vanish, corresponding to the three translations and three rotations along which V does not vary.

Whether there are exactly six is a property of the network, not a given. A displacement $\mathbf{u}$ is a null direction of $\mathbf{K}$ when it changes no active distance to first order, that is when $\hat{\mathbf{e}}_{ij} \cdot (\mathbf{u}_i - \mathbf{u}_j) = 0$ for every active spring. A network for which the only such displacements are the six rigid motions is *infinitesimally rigid*, and then $\text{rank } \mathbf{K} = 3N - 6$. Connectivity does not suffice: a node joined to the rest by a single active spring can swing about it, and one joined by two can rotate about the axis through them, in both cases without changing any active distance. Each such freedom is a further null direction. In what follows the networks are assumed infinitesimally rigid, and we write $\sum'$ and $\prod'$ for sums and products over the $3N - 6$ nonzero eigenvalues.

The distribution (23) is flat along the six null directions, which is the rigid-body freedom of the whole network, and we factor it out: displacements are taken in the complement, where the covariance is

$$\boldsymbol{\Sigma} = \beta^{-1} \mathbf{K}^+, \quad \mathbf{K}^+ = \sum_n' \lambda_n^{-1} \mathbf{q}_n \mathbf{q}_n^\top, \quad (24)$$

with $\mathbf{K}^+$ the pseudo-inverse obtained by inverting the nonzero eigenvalues and annihilating the null directions.

5 Thermodynamic properties

5.1 The partition function

In the quadratic approximation (22) the integral (14) factorises in the eigenbasis of $\mathbf{K}$. Writing $\mathbf{u} = \sum_n y_n \mathbf{q}_n$, the quadratic form is $\sum_n \lambda_n y_n^2$, and the integral over each of the $3N - 6$ non-null

directions is a standard Gaussian,

$$\int_{-\infty}^{\infty} e^{-\beta \lambda_n y_n^2 / 2} dy_n = \left(\frac{2\pi}{\beta \lambda_n} \right)^{1/2}. \quad (25)$$

Hence

$$Z = e^{-\beta V(\mathbf{r}^e; \Lambda)} \prod_n' \left(\frac{2\pi}{\beta \lambda_n} \right)^{1/2}, \quad \ln Z = -\beta V(\mathbf{r}^e; \Lambda) + \frac{1}{2} \sum_n' \ln \frac{2\pi}{\beta \lambda_n}. \quad (26)$$

5.2 Free energy, internal energy, entropy

The Helmholtz free energy is

$$A = -\beta^{-1} \ln Z = V(\mathbf{r}^e; \Lambda) - \frac{1}{2\beta} \sum_n' \ln \frac{2\pi}{\beta \lambda_n}. \quad (27)$$

The internal energy follows by differentiating (26), using $\partial \ln \lambda_n / \partial \beta = 0$,

$$U = -\frac{\partial \ln Z}{\partial \beta} = V(\mathbf{r}^e; \Lambda) + \frac{3N - 6}{2\beta}, \quad (28)$$

which is the minimum energy plus $\frac{1}{2} k_B T$ per non-rigid degree of freedom, as equipartition requires. The entropy is $S = (U - A)/T$, so

$$TS = U - A = \frac{1}{2\beta} \sum_n' \left[\ln \frac{2\pi}{\beta \lambda_n} + 1 \right]. \quad (29)$$

Soft directions, those of small λ_n , contribute most: the network gains entropy by being easy to deform.

Two comments on (27) and (29). The logarithm takes a dimensional argument, so the numerical values of A and TS depend on the units chosen for length, and A may fall below $V(\mathbf{r}^e; \Lambda)$. Neither is meaningful alone. What is meaningful is a difference between two networks at the same temperature, where the units cancel; those are the quantities of Section 7. The same applies to the momentum factor dropped in (14), and to the rigid-body volume factored out of (24): both are common to the two networks and cancel.

Open question. Equations (27) and (29) rest on the quadratic truncation (22). How good it is for a frustrated network, where the transverse term makes $\mathbf{K}$ differ from the standard form, has not been examined here.

6 Reference and mutant

6.1 The reference network

Everything above applies to any network. We now fix one, the *reference*, with parameters $\Lambda^{\text{ref}} = \{l_{ij}^{\text{ref}}\}$ and potential

$$V_{\text{ref}}(\mathbf{r}) \equiv V(\mathbf{r}; \Lambda^{\text{ref}}) = \frac{1}{2} \sum_{i < j} k(l_{ij}^{\text{ref}}) \left(d_{ij}(\mathbf{r}) - l_{ij}^{\text{ref}} \right)^2. \quad (30)$$

Write $k_{ij}^{\text{ref}} = k(l_{ij}^{\text{ref}})$ for its force constants, $\mathbf{r}_{\text{ref}}^e$ for an equilibrium conformation obtained from (10), and

$$s_{ij} = d_{ij}(\mathbf{r}_{\text{ref}}^e) - l_{ij}^{\text{ref}} \quad (31)$$

for its strains. Its Hessian $\mathbf{K}^{\text{ref}}$, spectrum $\{\lambda_n^{\text{ref}}\}$, and thermodynamic properties A_{ref} , U_{ref} , S_{ref} follow from Sections 4 and 5. The reference may be frustrated or not; nothing below assumes either.

6.2 Mutation

A mutation at site m perturbs the equilibrium lengths of the springs that connect m to the rest of the network, and leaves all others unchanged:

$$l_{ij}^{\text{mut}} = l_{ij}^{\text{ref}} + \delta l_{ij}, \quad \delta l_{ij} = 0 \text{ unless } i = m \text{ or } j = m. \quad (32)$$

The perturbations δl_{ij} are drawn from a distribution with zero mean and variance σ^2 , independently for each affected spring; σ sets the size of the mutation.

The mutant potential is (6) evaluated with these parameters,

$$V_{\text{mut}}(\mathbf{r}) \equiv V(\mathbf{r}; \Lambda^{\text{mut}}) = \frac{1}{2} \sum_{i < j} k(l_{ij}^{\text{ref}} + \delta l_{ij}) \left(d_{ij}(\mathbf{r}) - l_{ij}^{\text{ref}} - \delta l_{ij} \right)^2. \quad (33)$$

Comparing (30) and (33): the mutation changes both the equilibrium lengths and, through (4), the force constants. Define

$$\delta k_{ij} \equiv k(l_{ij}^{\text{ref}} + \delta l_{ij}) - k(l_{ij}^{\text{ref}}), \quad \text{so} \quad k_{ij}^{\text{mut}} = k_{ij}^{\text{ref}} + \delta k_{ij}. \quad (34)$$

This is an exact difference of two values of k , not a differential: no assumption is made that δl_{ij} is small or that k is differentiable. A spring whose equilibrium length crosses from a region where $k > 0$ to one where $k = 0$, or the reverse, has δk_{ij} of the size of k itself. Contacts are created and destroyed here, in the parameters, and not by any change in conformation.

This is why every spring carries an equilibrium length, the inactive ones included. An inactive spring with no l_{ij} could never acquire one inside the range where $k > 0$, so a mutation could only destroy contacts and never form them; carrying the parameter regardless lets (32) move a spring in either direction across that range.

The mutant has its own equilibrium conformation $\mathbf{r}_{\text{mut}}^e$, obtained from (10) with the mutant parameters, its own strains $s_{ij}^{\text{mut}} = d_{ij}(\mathbf{r}_{\text{mut}}^e) - l_{ij}^{\text{mut}}$, its own Hessian $\mathbf{K}^{\text{mut}}$ and spectrum $\{\lambda_n^{\text{mut}}\}$, and its own A_{mut} , U_{mut} , S_{mut} . Note that $\mathbf{r}_{\text{mut}}^e$ is obtained by minimising (33) itself, not by displacing $\mathbf{r}_{\text{ref}}^e$ along a response to the perturbation.

Even if the reference is unfrustrated, the mutant will in general be frustrated: the perturbed lengths l_{ij}^{mut} need not be simultaneously realisable by any conformation, so the mutant settles at a compromise with $s_{ij}^{\text{mut}} \neq 0$.

7 Mutational changes

7.1 Three energy differences

Two networks and two conformations give three energy differences worth naming. The first is the cost of changing the parameters at fixed conformation,

$$\Delta V_{\text{stress}} = V_{\text{mut}}(\mathbf{r}_{\text{ref}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e), \quad (35)$$

the second the energy released as the mutant relaxes to its own equilibrium,

$$\Delta V_{\text{relax}} = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{mut}}(\mathbf{r}_{\text{ref}}^e), \quad (36)$$

and their sum is the difference between the two minima,

$$\Delta V_{\text{min}} = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e) = \Delta V_{\text{stress}} + \Delta V_{\text{relax}}. \quad (37)$$

Since $\mathbf{r}_{\text{mut}}^e$ minimises V_{mut} , we have $V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) \leq V_{\text{mut}}(\mathbf{r})$ for every $\mathbf{r}$, and in particular for $\mathbf{r} = \mathbf{r}_{\text{ref}}^e$, so

$$\Delta V_{\text{relax}} \leq 0. \quad (38)$$

7.2 Decomposition of the stress energy

ΔV_{stress} can be written out in closed form. We do so in steps.

Step 1: separate the springs. Both (30) and (33) are sums over the same index set, all $\frac{1}{2}N(N-1)$ springs, so their difference is a sum of one term per spring,

$$\Delta V_{\text{stress}} = \sum_{i < j} \left[\frac{1}{2} k_{ij}^{\text{mut}} (d_{ij}^e - l_{ij}^{\text{mut}})^2 - \frac{1}{2} k_{ij}^{\text{ref}} (d_{ij}^e - l_{ij}^{\text{ref}})^2 \right], \quad d_{ij}^e \equiv d_{ij}(\mathbf{r}_{\text{ref}}^e). \quad (39)$$

This is where carrying a parameter for every spring, active or not, does its work: no spring appears in one sum and not the other, and a contact created or destroyed by the mutation is carried by δk_{ij} rather than by a change in the range of summation.

Step 2: eliminate the distance. By the definition of the reference strain (31), $d_{ij}^e = l_{ij}^{\text{ref}} + s_{ij}$, so the two brackets become

$$d_{ij}^e - l_{ij}^{\text{ref}} = s_{ij}, \quad d_{ij}^e - l_{ij}^{\text{mut}} = s_{ij} - \delta l_{ij}. \quad (40)$$

The conformation has now disappeared from the problem: what remains is expressed entirely in the reference strains and the perturbation.

Step 3: substitute the force constants. Using (34), the term for one spring is

$$\frac{1}{2} (k_{ij}^{\text{ref}} + \delta k_{ij}) (s_{ij} - \delta l_{ij})^2 - \frac{1}{2} k_{ij}^{\text{ref}} s_{ij}^2. \quad (41)$$

Step 4: split off the change in stiffness. Grouping the two terms in k_{ij}^{ref} and keeping the δk_{ij} term apart,

$$= \frac{1}{2} k_{ij}^{\text{ref}} \left[(s_{ij} - \delta l_{ij})^2 - s_{ij}^2 \right] + \frac{1}{2} \delta k_{ij} (s_{ij} - \delta l_{ij})^2. \quad (42)$$

The second piece is already in final form. No expansion has been made in obtaining it, and none is possible: δk_{ij} is not assumed small.

Step 5: expand the square. The bracket is a difference of two squares,

$$(s_{ij} - \delta l_{ij})^2 - s_{ij}^2 = -2 s_{ij} \delta l_{ij} + \delta l_{ij}^2, \quad (43)$$

which is a finite algebraic identity, not a truncated expansion: it is exact for any δl_{ij} .

Step 6: collect. Summing over springs,

$$\Delta V_{\text{stress}} = \underbrace{\frac{1}{2} \sum_{i < j} k_{ij}^{\text{ref}} \delta l_{ij}^2}_{(a)} - \underbrace{\sum_{i < j} k_{ij}^{\text{ref}} s_{ij} \delta l_{ij}}_{(b)} + \underbrace{\frac{1}{2} \sum_{i < j} \delta k_{ij} (s_{ij} - \delta l_{ij})^2}_{(c)}. \quad (44)$$

This holds for any k , any mutation size, and any reference network, frustrated or not.

7.3 The three terms

Term (a) comes from the δl_{ij}^2 of the expanded square. It is non-negative, quadratic in the size of the mutation, and independent of the state of the reference.

Term (b) is the cross term $-2s_{ij}\delta l_{ij}$ of the same square. It is proportional to the reference strain and is the only term of the three that can be negative. It contributes $-k_{ij}^{\text{ref}} s_{ij} \delta l_{ij}$ per spring, which is negative when s_{ij} and δl_{ij} share a sign: lengthening the equilibrium length of a stretched spring ($s_{ij} > 0$, $\delta l_{ij} > 0$), or shortening that of a compressed one, moves the equilibrium length towards the distance the spring actually spans and so releases energy. The opposite pairing costs energy. Term (b) vanishes when $s_{ij} \delta l_{ij} = 0$ for every spring, which is to say when the reference is unstrained on the springs the mutation touches. In particular it vanishes for any mutation of an unfrustrated reference, and it does not vanish in general.

Term (c) is the whole of the contribution from the change in force constants. It vanishes when $\delta k_{ij}(s_{ij} - \delta l_{ij})^2 = 0$ for every spring, in particular when k is constant over the range the mutation explores. It is the term that a model with a fixed contact map does not have.

Terms (b) and (c) are what distinguish (44) from the energy cost of mutating an unfrustrated network with fixed force constants, which is (a) alone. Neither is a correction of higher order in the mutation size: both are exact contributions of the same standing as (a), which happen to vanish in that case.

7.4 Differences and energies

ΔV_{stress} is a difference of two potentials, and is not the same as the absolute energy $V_{\text{mut}}(\mathbf{r}_{\text{ref}}^e)$ of the mutant network at the reference conformation. By (35) the two differ by $V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e)$, the strain energy the reference already carries, which by (13) is zero for an unfrustrated reference and positive otherwise.

7.5 Changes in the thermodynamic properties

Applying Section 5 to both networks and subtracting, the internal energy difference follows from (28),

$$\Delta U = U_{\text{mut}} - U_{\text{ref}} = V_{\text{mut}}(\mathbf{r}_{\text{mut}}^e) - V_{\text{ref}}(\mathbf{r}_{\text{ref}}^e) = \Delta V_{\text{min}}, \quad (45)$$

the equipartition terms cancelling. This step, and the entropy difference below, require the two networks to have the same number of non-null directions. A point mutation changes the parameters of one site and neither adds nor removes a node, so both Hessians are $3N \times 3N$; and if both networks are infinitesimally rigid in the sense of Section 4, both have $3N - 6$ nonzero eigenvalues. For the entropy, write (29) as

$$TS = \frac{1}{2\beta} \left[(3N - 6) \left(\ln \frac{2\pi}{\beta} + 1 \right) - \sum_n' \ln \lambda_n \right], \quad (46)$$

in which only the last term depends on the network. Both networks have the same number of non-null directions, so on subtracting, the constant term cancels and

$$T\Delta S = \frac{1}{2\beta} \left[\sum_n' \ln \lambda_n^{\text{ref}} - \sum_n' \ln \lambda_n^{\text{mut}} \right] = \frac{1}{2\beta} \ln \frac{\prod_n' \lambda_n^{\text{ref}}}{\prod_n' \lambda_n^{\text{mut}}}. \quad (47)$$

The two spectra enter only through these products. No correspondence between the modes of the reference and those of the mutant is used, and none is available: the eigenvectors of $\mathbf{K}^{\text{ref}}$ and $\mathbf{K}^{\text{mut}}$ are different sets of directions, and the eigenvalues are not paired by any property of the model.

The free energy difference is then

$$\Delta A = \Delta U - T\Delta S = \Delta V_{\min} - \frac{1}{2\beta} \ln \frac{\prod_n' \lambda_n^{\text{ref}}}{\prod_n' \lambda_n^{\text{mut}}}. \quad (48)$$

The dimensional ambiguity noted after (29) has cancelled: (47) and (48) depend only on the ratio of the two products. In the protein literature the quantity (48) is what is meant by $\Delta\Delta G$ for the folded state.

The products $\prod_n' \lambda_n$ are the pseudo-determinants of the two Hessians, so (47) may be computed without diagonalising either matrix.

Equations (37) and (47) separate the two ways a mutation changes the free energy. It shifts the minimum of the potential, which (44) decomposes; and it changes the spectrum of $\mathbf{K}$, which by (19) depends on the strains through g_{ij} and on the contact map through $k(l_{ij})$. A mutation that leaves the minimum energy untouched can still change the free energy, through the spectrum alone.

8 The step function

The results above hold for any k . The common choice is a step at a cutoff distance,

$$k(l) = k_0 \mathbb{1}[l \leq d_{\max}], \quad (49)$$

under which a spring is either active with strength k_0 or inactive. Two consequences follow from the results above.

First, V is discontinuous in the parameters. If a mutation carries l_{ij}^{ref} across $d_{\max}$ then $\delta k_{ij} = \pm k_0$, and by term (c) of (44) the energy changes by $\pm \frac{1}{2} k_0 (s_{ij} - \delta l_{ij})^2$ however small δl_{ij} is.

Second, and for the same reason, term (c) cannot be treated as small when δl_{ij} is small. This is the sense in which the contact map resists a perturbative treatment: δk_{ij} is not controlled by δl_{ij} .

A continuous k , for instance a sigmoid of width w centred on $d_{\max}$, makes V continuous in Λ and δk_{ij} small whenever δl_{ij} is small. The cost is that no spring is exactly inactive, so the contact map becomes a threshold imposed on a continuum rather than a property of the network.

9 Open questions and work to come

The following are not settled here.

- The quadratic truncation (22) is the only approximation in the document. Its accuracy for a frustrated network has not been examined.
- The magnitudes of (a), (b) and (c) in (44) for a real protein are not known. The decomposition says when each vanishes, not how large each is.

- How large $|s_{ij}|$ may become before a compressed spring ($s_{ij} < 0$), with its negative transverse stiffness (21), ceases to represent a residue contact.
- A mutation can cost the network its infinitesimal rigidity. The perturbation (32) acts on every spring of the mutated site, and if enough of their equilibrium lengths leave the range where $k > 0$, the site is left attached by one or two springs and acquires a rotational freedom that changes no active distance. The mutant Hessian then has more than six null directions, the counts in Section 7 no longer match, and neither $\Delta U = \Delta V_{\min}$ nor (47) holds as written: an uncanceled term in $\ln(2\pi/\beta)$ survives, so ΔA is no longer independent of the units of length. How often this happens at a given mutation size and cutoff is an empirical question, and what the free energy difference should mean when it does is not settled here.
- How the parameters Λ should be chosen. Section 3.3 gives one recipe, $l_{ij} = d_{ij}(\mathbf{r}^*)$ for a chosen structure, which yields an unfrustrated network; but nothing selects it, and the question of whether a frustrated network is the better model of a native structure — and if so, which one — is open. This is a question about the parameters, and is not one (10) addresses: that equation takes Λ as given and solves for the conformation.

To be added: the distribution of structures generated by repeated mutation, and the entropic contribution to $\Delta\Delta G$ along a substitution process.